EXPERIMENT: 10

PERFORMANCE EVALUATION OF DIGITAL COMMUNICATION

SYSTEMS USING EYE DIAGRAM ANALYSIS UNDER ADDITIVE

WHITE GAUSSIAN NOISE (AWGN) USING MATLAB

AIM:

1. To generate a digital communication signal in MATLAB and transmit it through an

Additive White Gaussian Noise (AWGN) channel.

2. To obtain and analyze the eye diagram of the received digital signal for different noise

conditions.

3. To measure eye-opening parameters such as eye height, eye width, crossing level and

timing variation.

4. To study the effect of decreasing SNR / Eb/N0 on inter-symbol interference, noise margin

and timing performance.

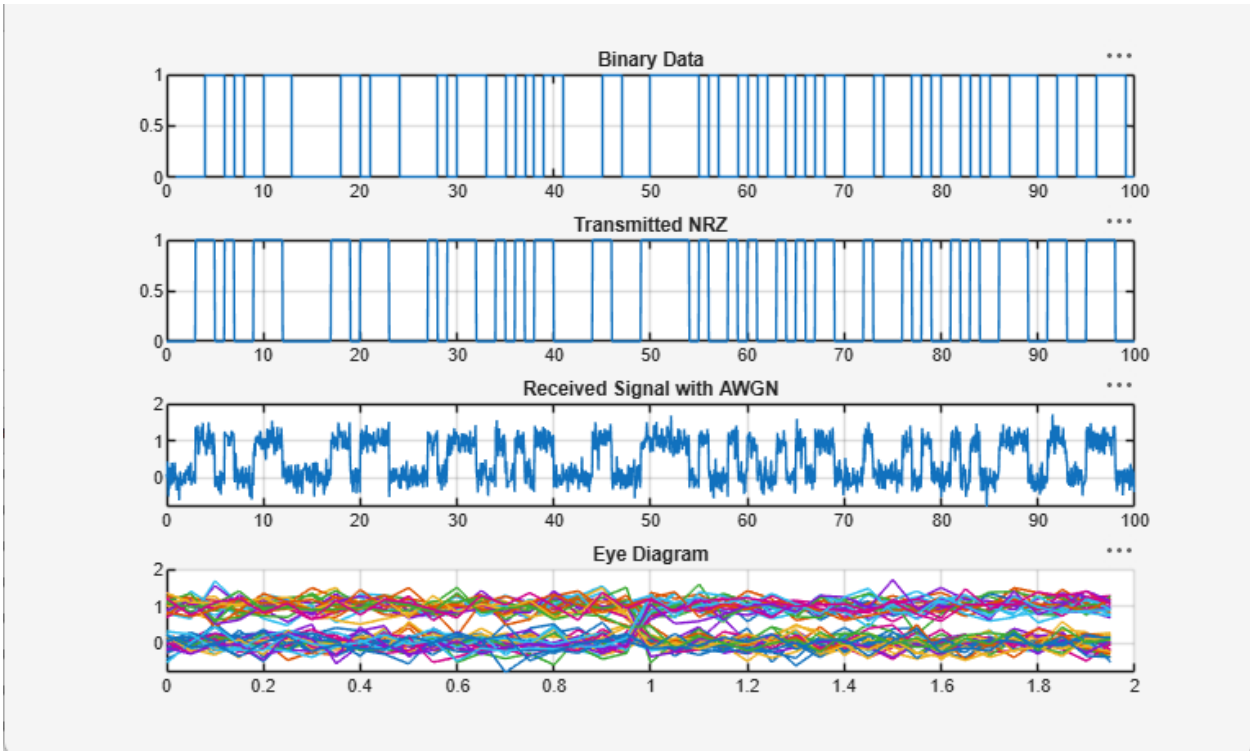
5. To compare the transmitted and received signal characteristics and evaluate

communication-system performance.

CODE:

```
clc; clear; close all;
N=100; sps=20; SNR=10;
data=randi([0 1],1,N);
tx=repelem(data,sps);
% Add AWGN manually
noise=sqrt(mean(tx.^2)/10^(SNR/10))*randn(size(tx));
rx=tx+noise;
t=(0:length(tx)-1)/sps;
figure;
subplot(4,1,1); stairs(data); title('Binary Data'); grid on;
subplot(4,1,2); plot(t,tx); title('Transmitted NRZ'); grid on;
subplot(4,1,3); plot(t,rx); title('Received Signal with AWGN'); grid on;
subplot(4,1,4); hold on;
for k=1:2*sps:length(rx)-2*sps+1
    plot((0:2*sps-1)/sps,rx(k:k+2*sps-1));
end
title('Eye Diagram'); grid on; hold off;
```

OUTPUT:



RESULT:

i) Successfully generated an NRZ digital communication signal and transmitted it through an AWGN channel.

ii) Observed that the received signal was affected by Gaussian noise, resulting in signal distortion and bit errors.

iii) Successfully plotted the eye diagram of the received signal using MATLAB.

iv) Observed that the eye opening decreased due to AWGN, indicating reduced signal quality and increased probability of bit errors.